

Total No. of Questions : 8]

SEAT No. :

PD4084

[Total No. of Pages : 2

[6402]-44

S.E. (Computer Engineering)

PRINCIPLES OF PROGRAMMING LANGUAGES

(2019 Pattern) (Semester - IV) (210255)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain why Java is secure, portable and Dynamic. Which of the concepts in Java ensures these? [8]
- b) What is the String class in Java? List and explain with proper examples any four methods of the String class. [9]

OR

- Q2)** a) Discuss the importance of garbage collection in Java. Explain how the finalize() method is related to garbage collection. Differentiate garbage collection in Java and C++? [8]
- b) What do you mean by method overloading? Demonstrate through a program in Java how method overloading is used to add two integers and two strings respectively. [9]

- Q3)** a) Describe the constructor call sequence in Java when dealing with inheritance. How does it differ for constructors of the superclass and subclass with proper examples? [9]
- b) What are interfaces in Java? Explain how interfaces are defined and implemented. Provide an example of extending interfaces. [9]

OR

- Q4)** a) Discuss the types of exceptions in Java. Explain the concepts of uncaught exceptions, try-catch blocks, throw, throws, and finally with examples. [9]
- b) Define streams in Java. Differentiate between Byte Streams and Character Streams. Explain the purpose of Predefined Streams. [9]

P.T.O.

Q5) a) Compare and contrast implementing threads using the Thread class and the Runnable interface in Java. Provide examples for both approaches. [8]

b) Introduce three popular JavaScript frameworks used in web development: ReactJS, VueJS, and AngularJS. Discuss their features and advantages. [9]

OR

Q6) a) Discuss the advantages and disadvantages of using threads in Java programs. Provide examples illustrating scenarios where threads are beneficial and where they may lead to issues. [8]

b) Explain the role of JavaScript in creating web-based applications using Java. Provide examples of how JavaScript can be integrated with Java applications. [9]

Q7) a) Discuss the concepts of definitions, predicates, conditionals, and scoping in Lisp. Provide code examples for each concept. [9]

b) Write a Lisp program to calculate the factorial of a given number using recursion. [9]

OR

Q8) a) Compare and contrast the Functional Programming Paradigm and the Logic Programming Paradigm. Highlight their differences in terms of syntax, semantics, and problem-solving approaches. [9]

b) Develop a Prolog program on two numbers for basic arithmetic operations (addition, subtraction, multiplication, division). [9]

